

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Display graphs
%matplotlib inline
```

```
# Graph style
sns.set_style("whitegrid")
```

```
df = pd.read_csv("train.csv")
```

```
df.head()
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

		Name	Sex	Age
SibSp	\			
0		Braund, Mr. Owen Harris	male	22.0
1				
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	
1				
2	Heikkinen, Miss. Laina	female	26.0	
0				
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	
1				
4	Allen, Mr. William Henry	male	35.0	
0				

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
df.tail()
```

	PassengerId	Survived	Pclass	
Name	\			
886	887	0	2	Montvila, Rev. Juozas
887	888	1	1	Graham, Miss. Margaret Edith
888	889	0	3	Johnston, Miss. Catherine Helen

```
"Carrie"
889      890      1      1      Behr, Mr. Karl
Howell
890      891      0      3      Dooley, Mr.
Patrick
```

	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	male	27.0	0	0	211536	13.00	NaN	S
887	female	19.0	0	0	112053	30.00	B42	S
888	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	male	26.0	0	0	111369	30.00	C148	C
890	male	32.0	0	0	370376	7.75	NaN	Q

```
df.sample(5)
```

	PassengerId	Survived	Pclass	
Name	Sex	\		
120	121	0	2	Hickman, Mr. Stanley
George	male			
54	55	0	1	Ostby, Mr. Engelhart
Cornelius	male			
802	803	1	1	Carter, Master. William Thornton
II	male			
261	262	1	3	Asplund, Master. Edvin Rojj
Felix	male			
532	533	0	3	Elias, Mr. Joseph
Jr	male			

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
120	21.0	2	0	S.O.C. 14879	73.5000	NaN	S
54	65.0	0	1	113509	61.9792	B30	C
802	11.0	1	2	113760	120.0000	B96 B98	S
261	3.0	4	2	347077	31.3875	NaN	S
532	17.0	1	1	2690	7.2292	NaN	C

```
df.shape
```

```
(891, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
```

```

5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
df.isnull().sum()
```

```

PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64

```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df = df.drop_duplicates()
df["Survived"].value_counts()

Survived
0      549
1      342
Name: count, dtype: int64

df["Sex"].value_counts()

Sex
male      577
female    314
Name: count, dtype: int64

df["Pclass"].value_counts()

Pclass
3      491
1      216
2      184
Name: count, dtype: int64

df["Embarked"].value_counts()

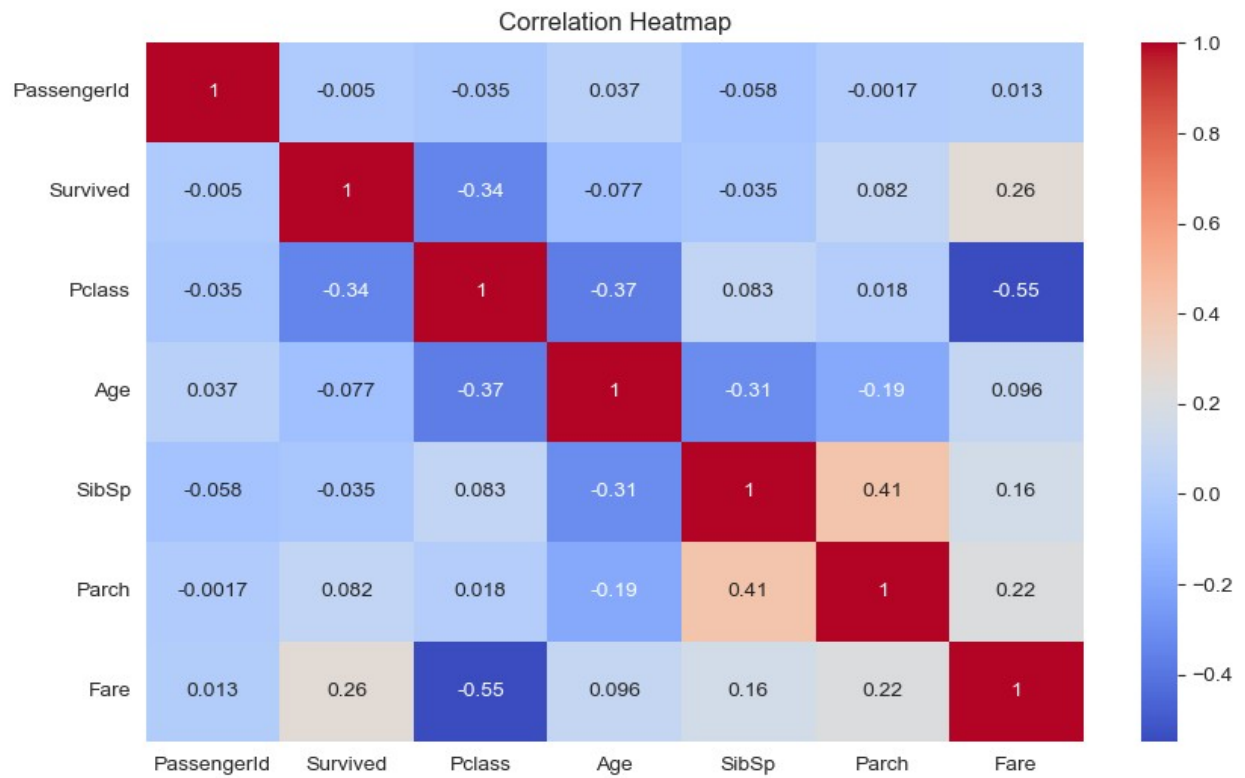
Embarked
S      644
C      168
Q       77
Name: count, dtype: int64

numeric_df = df.select_dtypes(include="number")

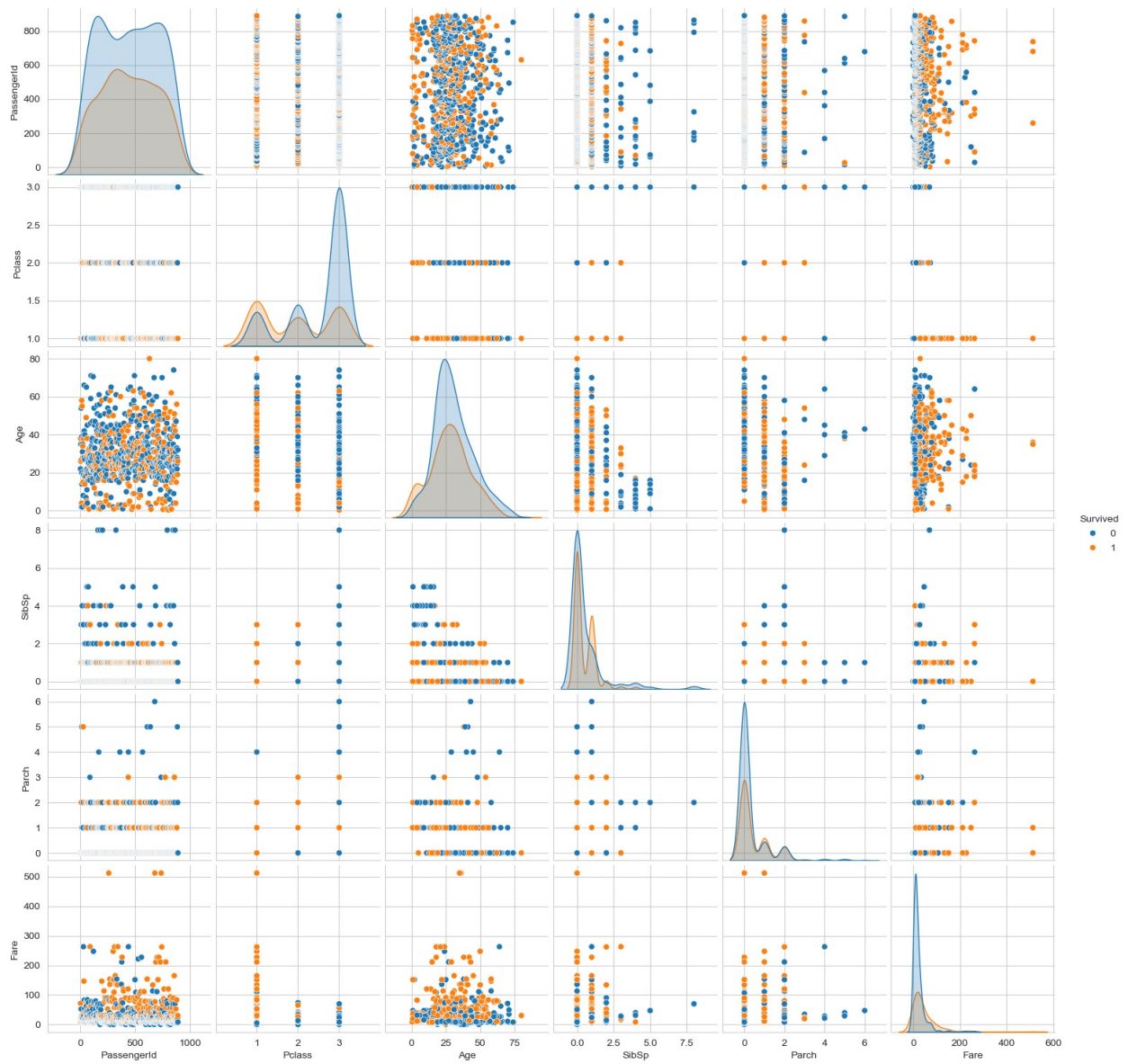
plt.figure(figsize=(10,6))

sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap="coolwarm")

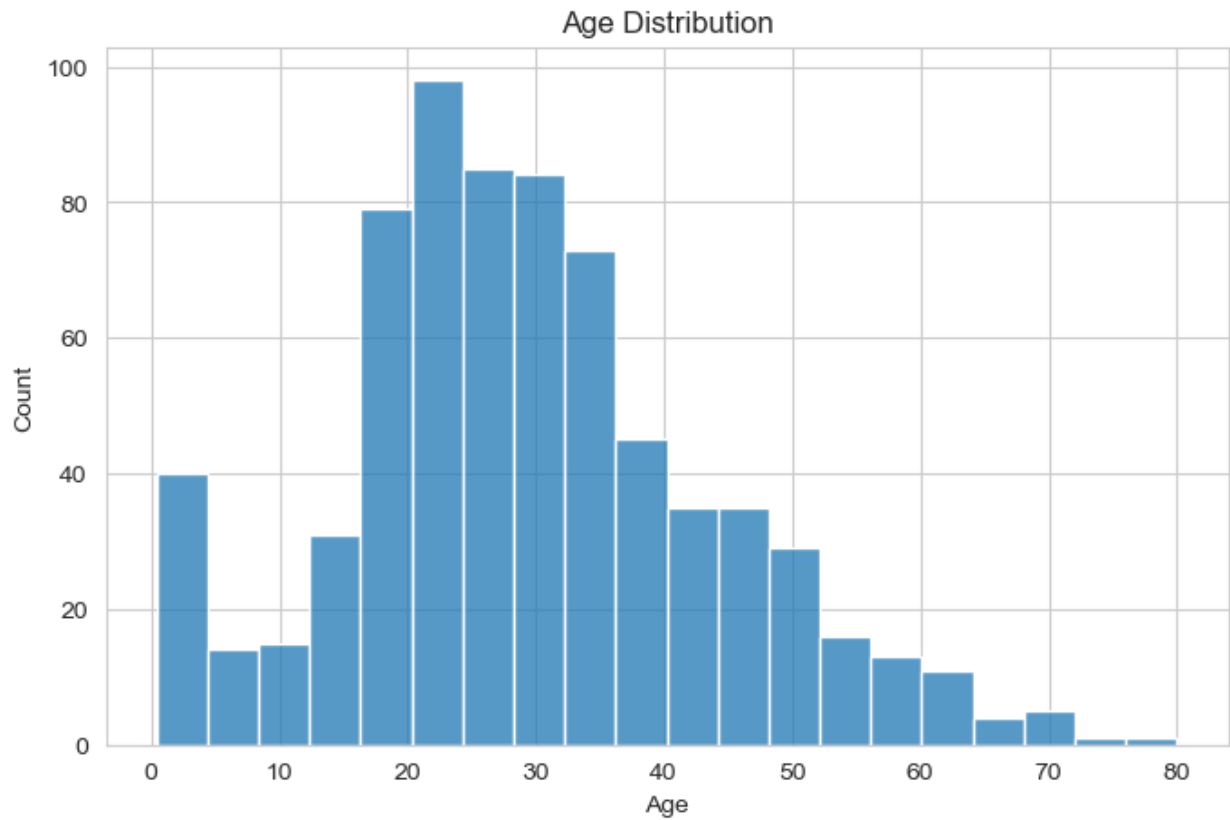
plt.title("Correlation Heatmap")
plt.show()
```



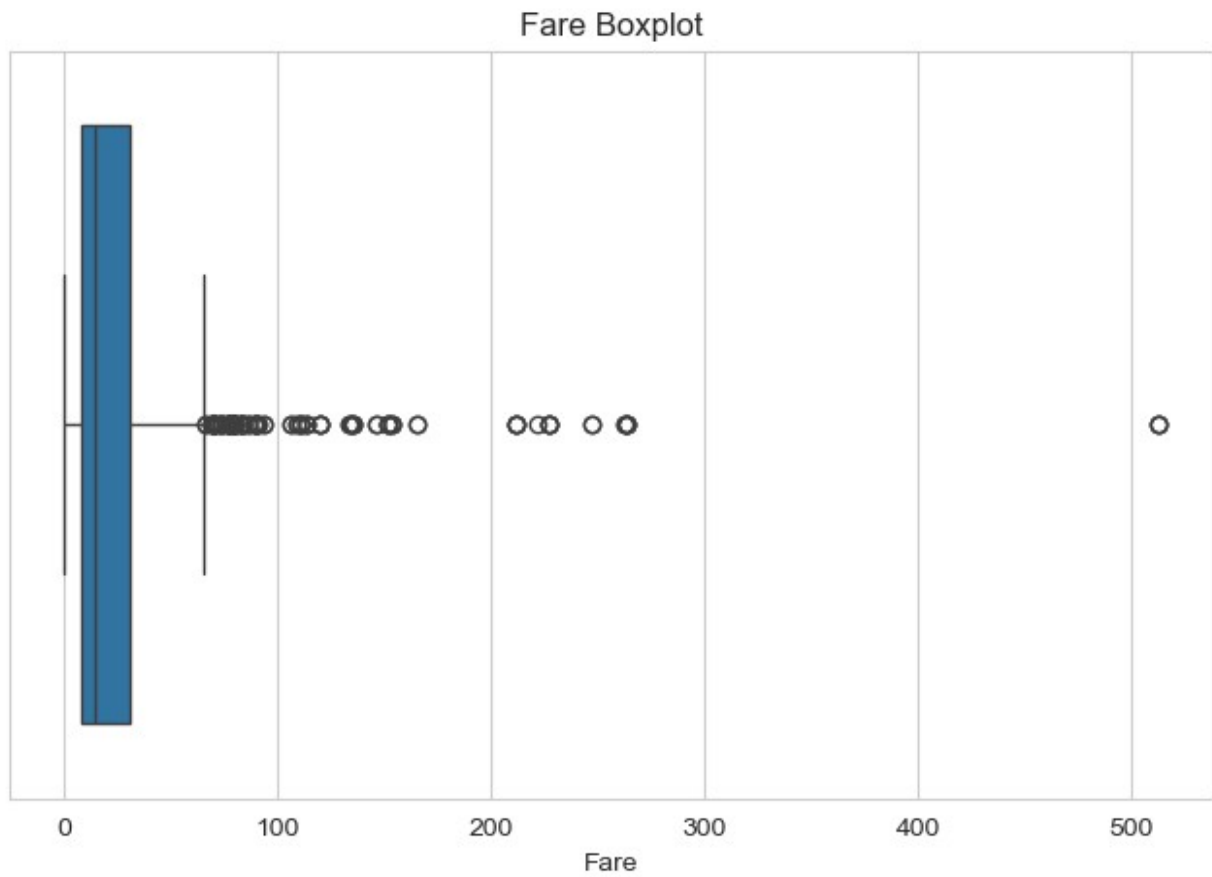
```
sns.pairplot(df, hue="Survived")  
plt.show()
```



```
plt.figure(figsize=(8,5))  
sns.histplot(df["Age"], bins=20)  
plt.title("Age Distribution")  
plt.show()
```



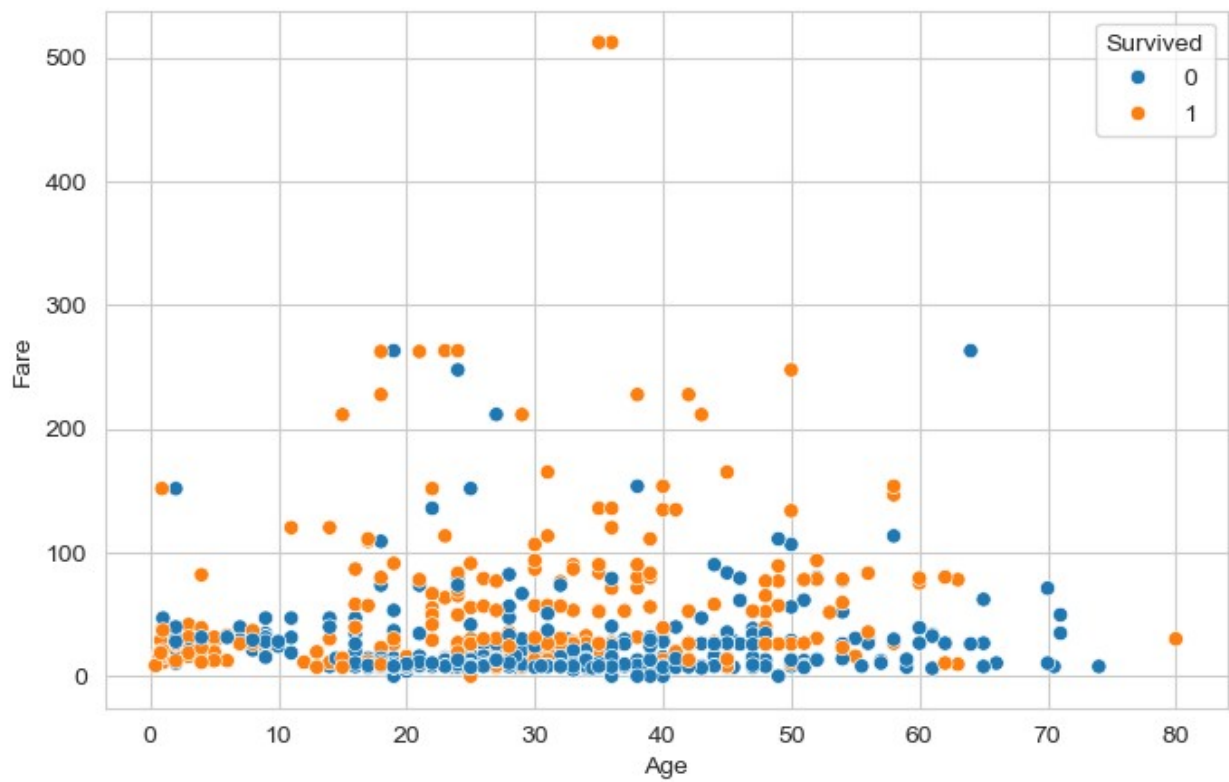
```
plt.figure(figsize=(8,5))
sns.boxplot(x=df["Fare"])
plt.title("Fare Boxplot")
plt.show()
```



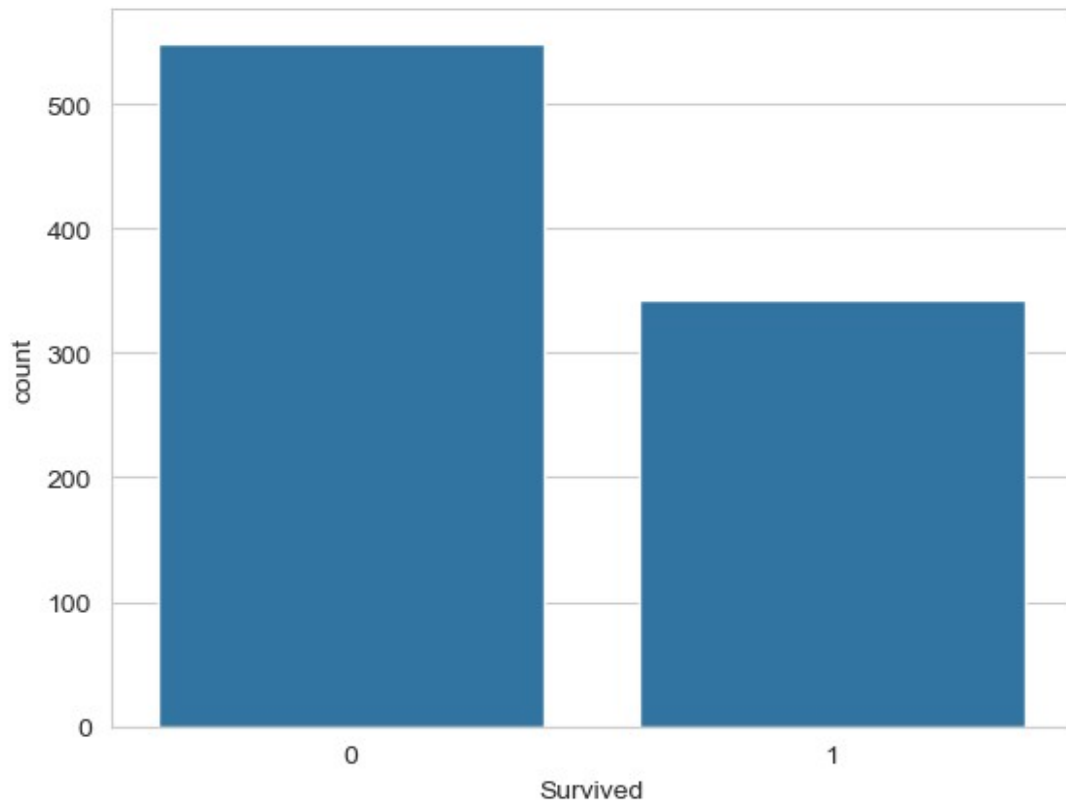
```
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x="Age",
    y="Fare",
    hue="Survived"
)

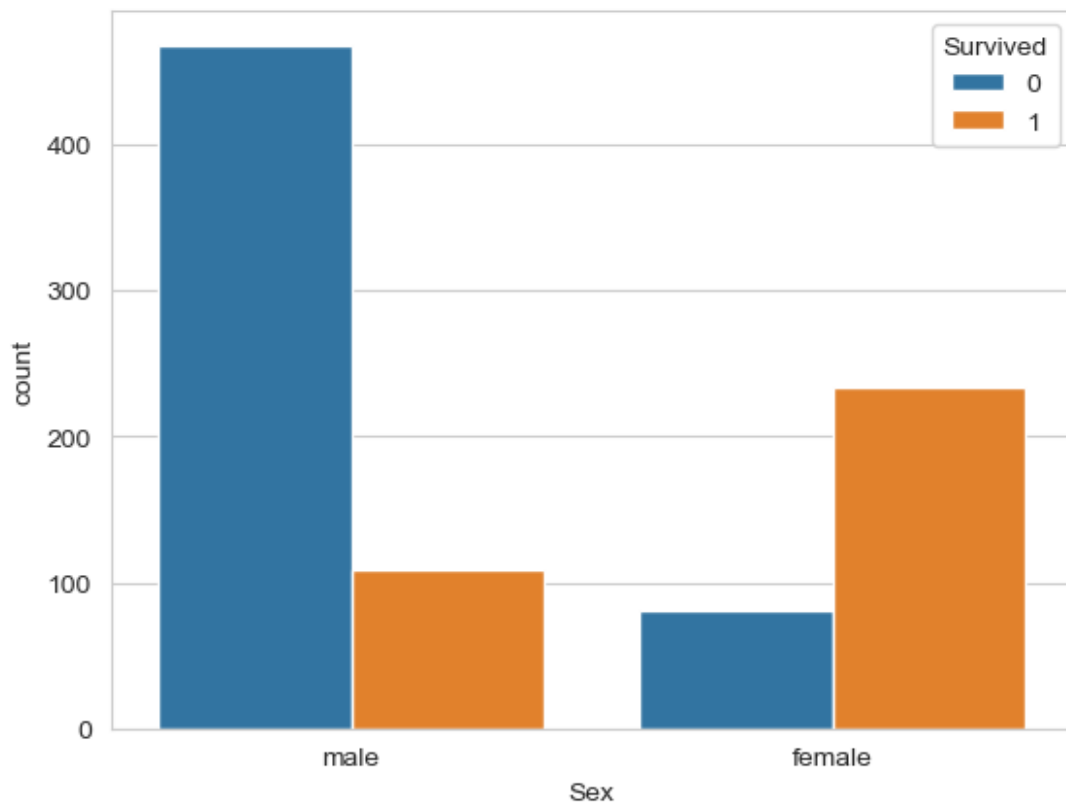
plt.show()
```

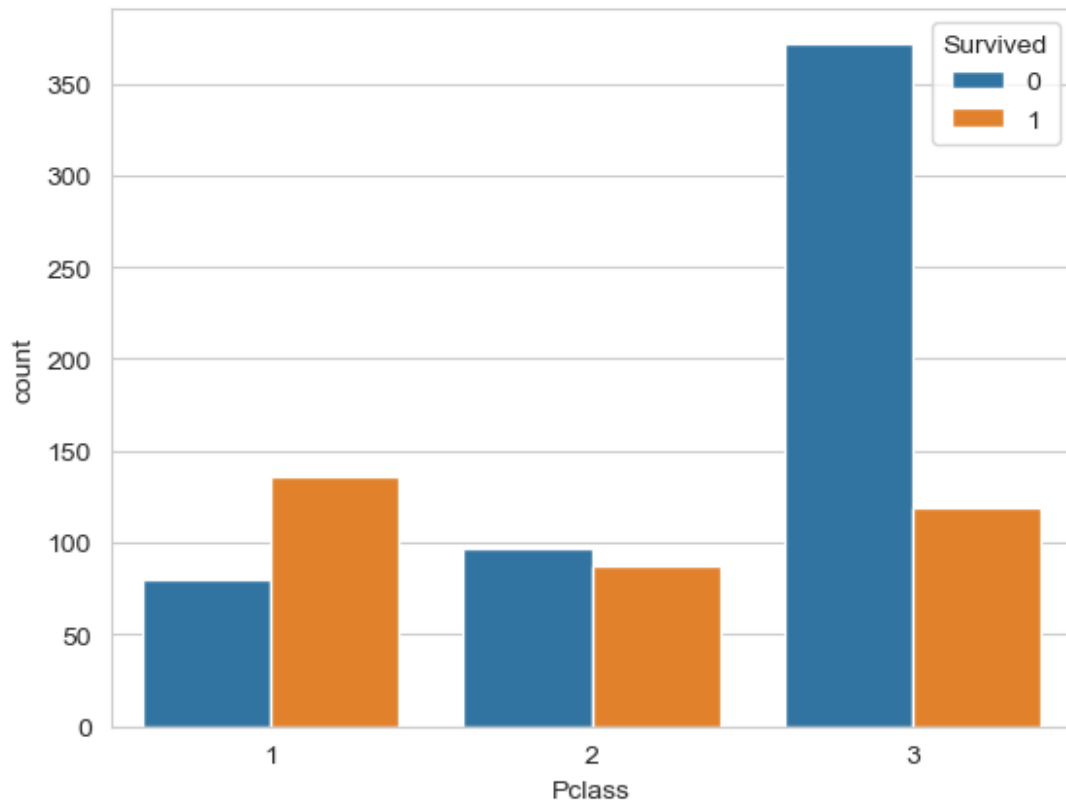
```
sns.countplot(  
    data=df,  
    x="Survived"  
)  
plt.show()
```



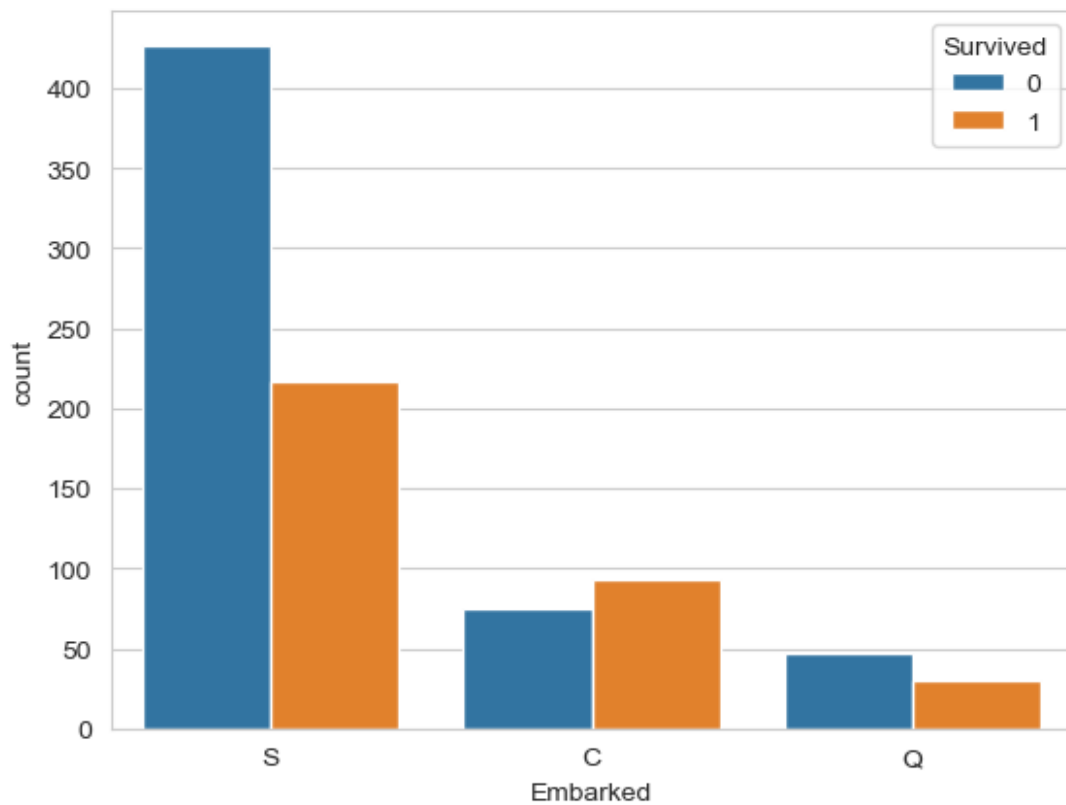
```
sns.countplot(  
    data=df,  
    x="Sex",  
    hue="Survived"  
)  
plt.show()
```



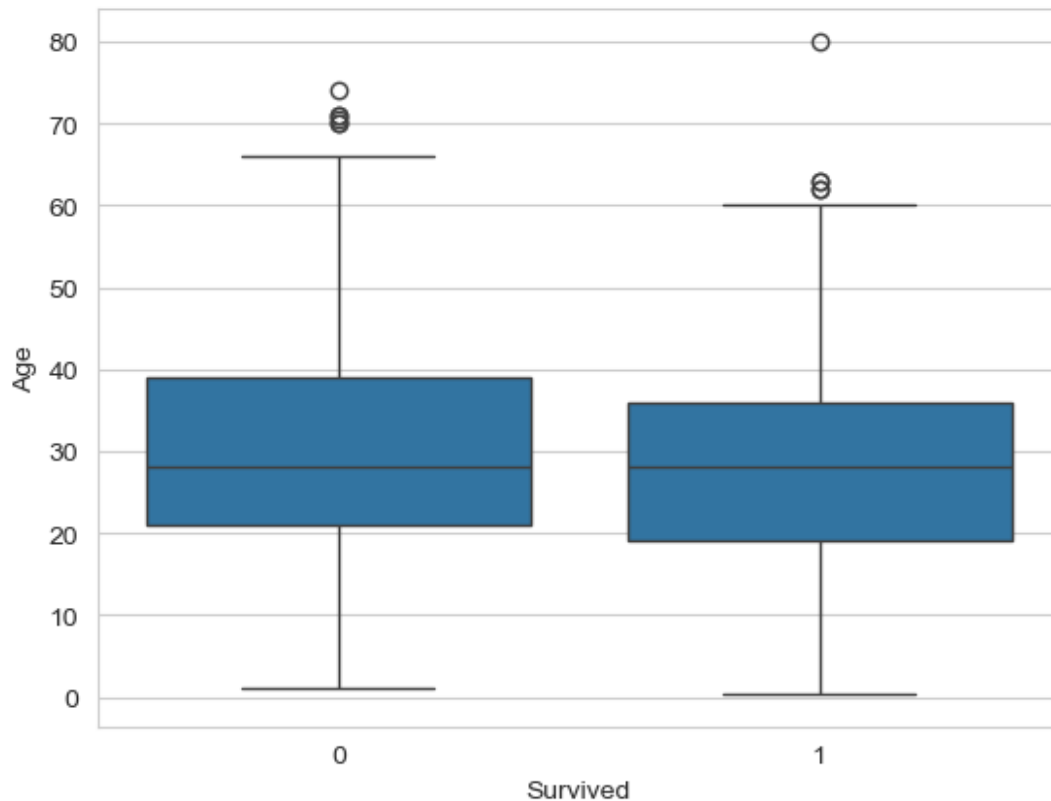
```
sns.countplot(  
    data=df,  
    x="Pclass",  
    hue="Survived"  
)  
plt.show()
```



```
sns.countplot(  
    data=df,  
    x="Embarked",  
    hue="Survived"  
)  
plt.show()
```



```
sns.boxplot(  
    data=df,  
    x="Survived",  
    y="Age"  
)  
plt.show()
```



```
=====
EXPLORATORY DATA ANALYSIS (EDA) OBSERVATIONS
Titanic Dataset
=====
```

1. Dataset Information (df.info())

Observations:

- The dataset contains 891 passenger records and 12 features.
- The Age, Cabin, and Embarked columns contain missing values.
- The dataset includes both numerical and categorical features.

2. Statistical Summary (df.describe())

Observations:

- The average passenger age is approximately 29.7 years.
- Passenger fares vary significantly across passengers.
- The maximum fare is much higher than the average fare, indicating outliers.

3. Missing Values (df.isnull().sum())

Observations:

- Cabin has the highest number of missing values.
 - Age contains several missing values.
 - Embarked has only a few missing values.
-

4. Survival Count Plot

Observations:

- Most passengers did **not** survive the disaster.
 - The number of deaths **is** higher than the number of survivors.
 - The dataset contains more non-survivors than survivors.
-

5. Gender Count Plot

Observations:

- Male passengers are more than female passengers.
 - Female passengers represent a smaller portion of the dataset.
 - The gender distribution **is not** balanced.
-

6. Gender vs Survival

Observations:

- Female passengers had a higher survival rate than male passengers.
 - Most male passengers did **not** survive.
 - Gender strongly influenced survival.
-

7. Passenger Class Count Plot

Observations:

- Third Class has the highest number of passengers.
 - First Class has the fewest passengers.
 - Most passengers traveled **in** Third Class.
-

8. Passenger Class vs Survival

Observations:

- First Class passengers had the highest survival rate.
- Third Class passengers had the highest number of deaths.
- Passenger **class** has a strong relationship **with** survival.

9. Embarked Count Plot

Observations:

- Most passengers boarded from Southampton (S).
 - Cherbourg (C) had fewer passengers.
 - Queenstown (Q) had the fewest passengers.
-

10. Embarked vs Survival

Observations:

- Survival rates differ by embarkation port.
 - Passengers from Cherbourg had a higher survival rate.
 - Southampton had the largest number of passengers.
-

11. Age Histogram

Observations:

- Most passengers were between 20 and 40 years old.
 - Very few passengers were older than 70 years.
 - The age distribution is slightly right-skewed.
-

12. Fare Histogram

Observations:

- Most passengers paid relatively low fares.
 - Only a few passengers paid very high fares.
 - The fare distribution is highly right-skewed.
-

13. Fare Boxplot

Observations:

- Several extreme fare values appear as outliers.
 - Most ticket fares are concentrated at lower values.
 - High-fare passengers mainly belong to First Class.
-

14. Age Boxplot

Observations:

- A few passengers have unusually high ages.

- Most passengers fall within the normal age range.
 - Some age values are considered outliers.
-

15. Scatter Plot (Age vs Fare)

Observations:

- There is no strong relationship between age and fare.
 - Higher fare passengers generally had better survival.
 - Fare has a stronger relationship with survival than age.
-

16. Pair Plot

Observations:

- Passenger fare and class show noticeable differences between survivors and non-survivors.
 - Age alone does not clearly separate survivors from non-survivors.
 - Survival depends on multiple factors.
-

17. Correlation Heatmap

Observations:

- Passenger Class is negatively correlated with Fare.
 - Survival has a positive correlation with Fare.
 - Most numerical variables have weak correlations.
-

18. Age vs Survival Boxplot

Observations:

- Survivors are slightly younger on average.
 - Children had better survival chances.
 - Age alone is not a strong predictor of survival.
-

19. SibSp Count Plot

Observations:

- Most passengers traveled without siblings or spouses.
 - Some passengers traveled with one or two family members.
 - Large family groups were uncommon.
-

20. Parch Count Plot

Observations:

- Most passengers traveled without parents **or** children.
- Few passengers traveled **with** family members.
- Large family groups were rare.

FINAL SUMMARY

- The Titanic dataset contains **891** passenger records **and** **12** features.
- Missing values are present mainly **in** the Age, Cabin, **and** Embarked columns.
- Most passengers traveled **in** Third Class.
- Female passengers had a much higher survival rate than male passengers.
- First Class passengers had better survival rates than Third Class passengers.
- Most passengers were between **20 and 40** years old.
- Ticket fares contain several high-value outliers.
- Passenger **class**, gender, **and** fare are the most important factors related to passenger survival.